

# Concurrency and Distributed Systems

## Week 9 – Part 2: Multiprocessing and Visual Proof

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# Part 2 Roadmap

1 Processes

2 Visual Proof

3 Demo

# Why Multiprocessing Changes the Story

- Multiple processes have separate Python interpreters.
- Separate interpreters can run on separate cores.
- Data must cross process boundaries.
- Startup, memory, and serialization costs matter.

# Cost Model

| Model     | Good for                   | Cost                           |
|-----------|----------------------------|--------------------------------|
| Threads   | I/O overlap, shared memory | GIL limits CPU-bound bytecode  |
| Processes | CPU-bound parallelism      | startup, memory, serialization |

# Proof Procedure

- ① Open Task Manager, `htop`, or `py-spy`.
- ② Run CPU-bound thread demo.
- ③ Run CPU-bound process demo.
- ④ Compare logical CPU utilization.
- ⑤ Record command, observation, and explanation.

# What to Look For

- CPU-bound threads: limited effective parallel execution.
- Multiprocessing: multiple interpreters can use multiple cores.
- Small examples may be dominated by overhead.
- The conclusion must mention workload and machine.

Run

```
.\run-demo.ps1 02
```

Students complete VISUAL\_PROOF\_GUIDE.md.